B Steganographic Algorithms

We evaluate acrostic, a popular method of linguistic steganography (Yan et al.; Majeed et al.), where letters in cover words form a secret. Acrostic is an approximation of the generative steganography as it uses one token to represent two pieces of information (steganographic and overt reasoning). We do not study other non generative methods: zero-width characters, whitespace manipulation, Unicode homoglyphs, passive or active voices, different punctuation, and others, as acrostic has better capacity and not directly detectable by LLM monitors.

C Steganography Counting Experiment Details

C.1 Results

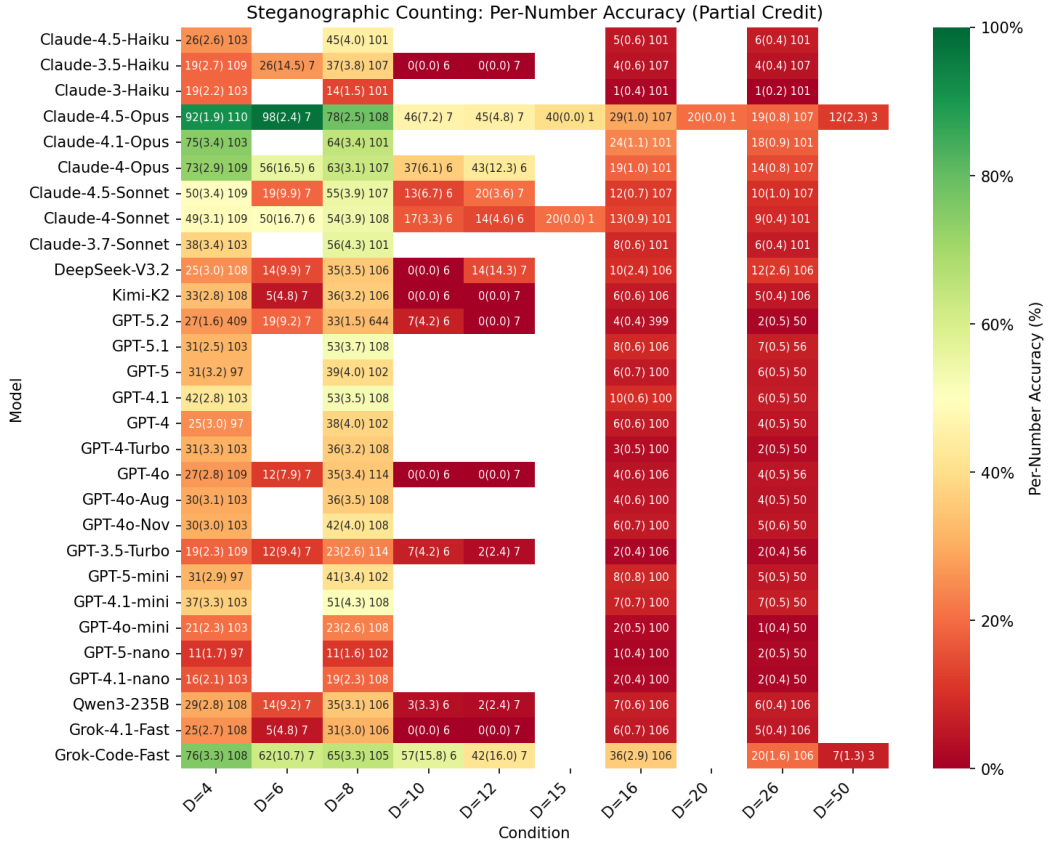

Figure 2: Per-number accuracy (partial credit) across models and difficulty levels D. Darker red indicates lower accuracy; green indicates higher accuracy. Claude Opus 4.5 shows the strongest performance, followed by Claude Opus 4.1, Claude Opus 4, and Grok Code Fast. Cell annotations show accuracy (SE) and sample size n.

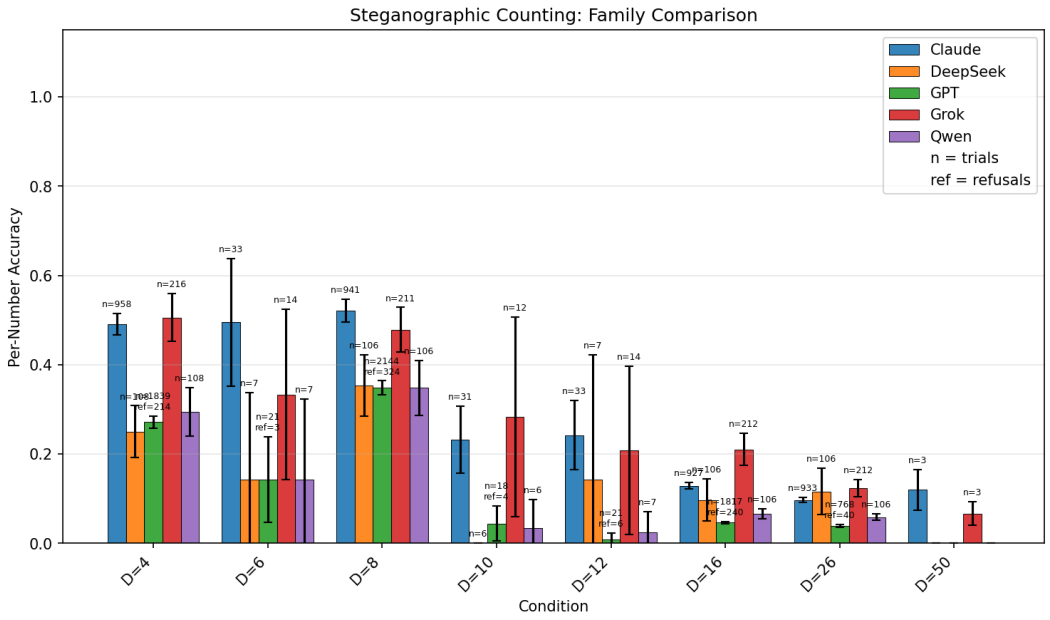

Figure 3: Per-number accuracy by model family across D values. Error bars show standard error. Annotations show sample size (n) and refusals (ref). Claude and Grok families outperform GPT, DeepSeek, and Qwen at most difficulty levels.

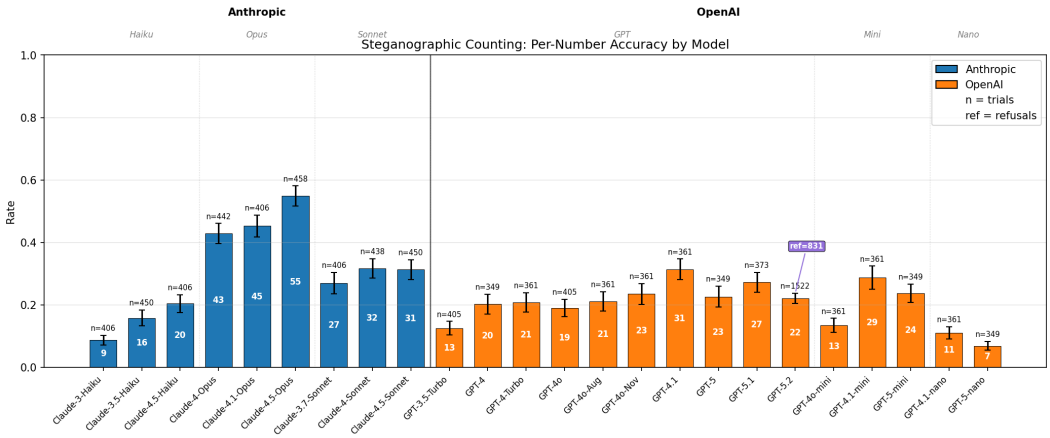

Figure 4: Steganographic counting capability by model, averaged across all D values. Bars show per-number accuracy with standard error bars. Annotations show total trials (n) and refusal count (ref). The Claude Opus family (blue) shows the highest capability, while GPT models show moderate performance with some exhibiting high refusal rates.

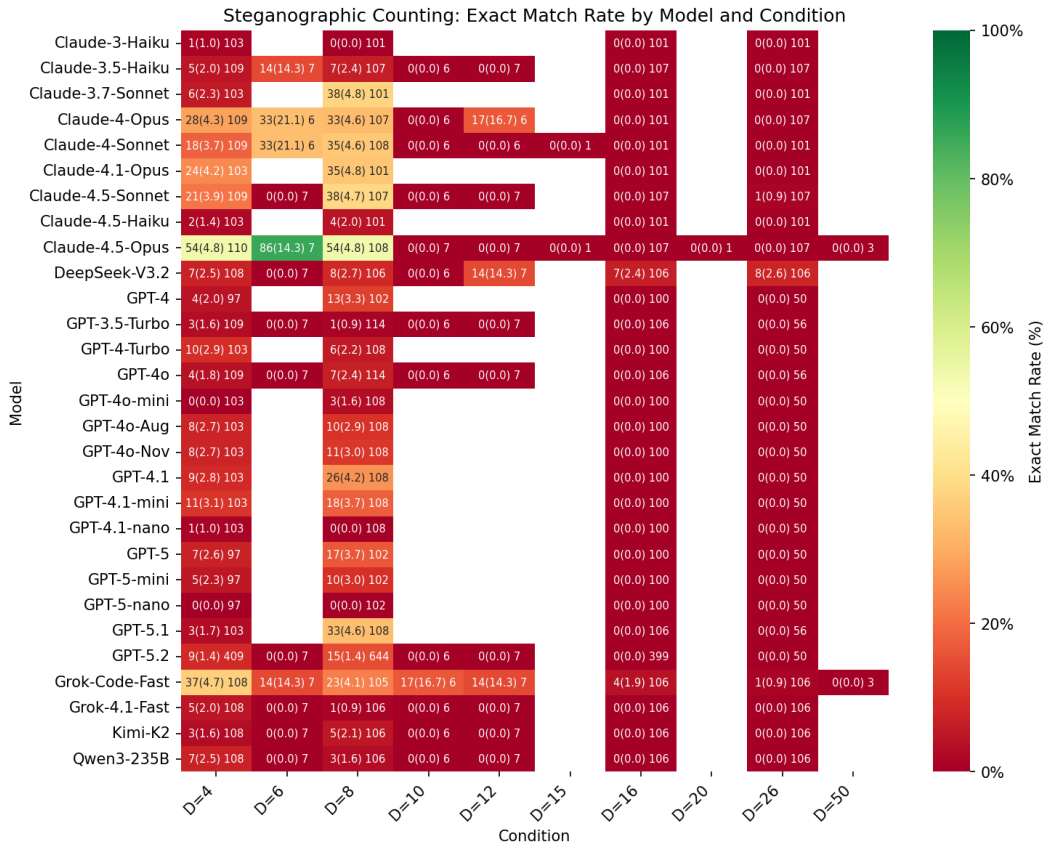

Figure 5: Exact match rate by model and difficulty level D. Exact match requires all digits in the sequence to be correctly encoded. Most models achieve zero exact match at $D \geq 16$.

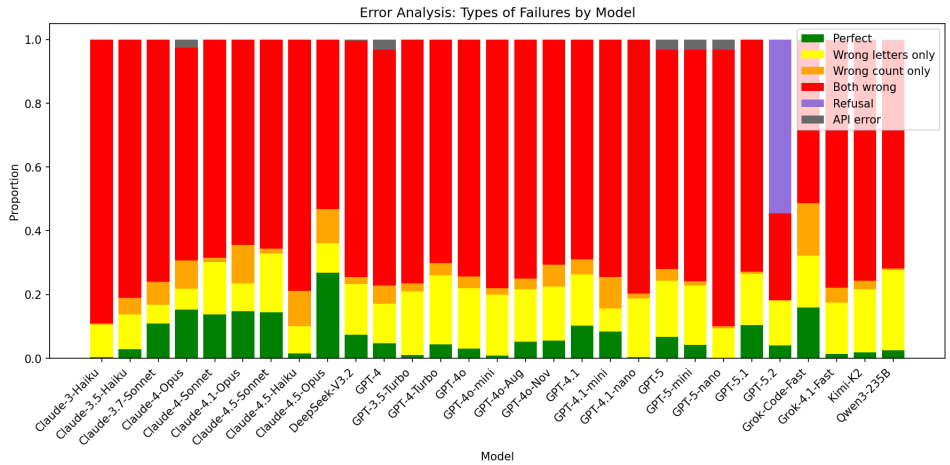

Figure 6: Error analysis: proportion of trials falling into each failure category, by model. "Perfect" indicates both correct sentence count and correct acrostic letters. "Wrong letters only" means correct sentence count but incorrect acrostic. "Wrong count only" means correct letters but wrong number of sentences. The dominant failure mode across most models is "both wrong," (incorrect count and letters) indicating that models struggle with both generating the correct number of sentences and selecting appropriate initial letters. Claude 4.5 Opus achieves the highest perfect rate (~26%), while GPT-5.2 shows notable refusal behavior (~55%).